Critical Analysis: Automatic Chain-of-Thought Prompting in Large Language Models

Kai-Yu Lu

1 Methodological Strengths

1.1 Comprehensive and structured experimental design

The paper adopts a broad and well organized experimental design across ten public reasoning benchmarks that cover three categories of tasks: arithmetic reasoning (MultiArith, GSM8K, AddSub, AQuA, SingleEq, SVAMP), commonsense reasoning (CSQA, StrategyQA), and symbolic reasoning (Last Letter Concatenation, Coin Flip). All evaluations use accuracy as the main metric, which is appropriate for single-answer reasoning tasks and allows direct comparison across baselines.

The experimental protocol is carefully aligned with prior work on Chain-of-Thought (CoT) prompting. The study evaluates five prompting paradigms using GPT-3 (text-davinci-002): Zero-Shot, Zero-Shot-CoT, Few-Shot, Manual-CoT, and the proposed Auto-CoT. Zero-Shot and Zero-Shot-CoT results are taken from Kojima et al. and Few-Shot and Manual-CoT results are taken from Wei et al. The Auto-CoT results are averaged over three random runs, which gives some measure of stability over stochastic demonstration construction. This consistent design ensures that performance gains can be attributed to the Auto-CoT mechanism instead of confounding differences in model or evaluation setup.

The main comparison in Table 3 shows that Auto-CoT matches or slightly outperforms Manual-CoT on most datasets. For example, on MultiArith, Auto-CoT achieves 92.0% accuracy compared to 91.7% for Manual-CoT and 78.7% for Zero-Shot-CoT. On GSM8K, Auto-CoT reaches 47.9%, improving over Manual-CoT at 46.9% and clearly surpassing Zero-Shot-CoT at 40.7%. On the symbolic Last Letter Concatenation task, Auto-CoT obtains 59.7%, slightly above 59.0% for Manual-CoT and much higher than the 0.2% of Zero-Shot and Few-Shot without CoT. This consistent pattern provides convincing evidence that automatically constructed CoT demonstrations can rival hand-crafted prompts.

The study further examines generality across models by replacing GPT-3 with Codex (code-davinci-002). Table 4 demonstrates that with Codex, Auto-CoT maintains competitive or superior performance relative to Manual-CoT. For example, on AddSub, Auto-CoT yields 91.9% while Manual-CoT obtains 84.6% and Zero-Shot-CoT 65.6%. This cross-model comparison strengthens the claim that the method is not tightly coupled to a single LLM.

1.2 Careful problem formulation and baseline analysis

The paper explicitly formulates the challenge of Auto-CoT: given only a set of questions without labels or human-written demonstrations, construct a small set of CoT demonstrations that can serve as in-context examples. Rather than directly proposing a solution, the authors first conduct an in-depth analysis of failure modes of a natural baseline, Retrieval-Q-CoT.

Retrieval-Q-CoT constructs demonstrations by retrieving the nearest neighbor questions to a test query in an embedding space and pairing them with Zero-Shot-CoT-generated rationales and answers. The study focuses on MultiArith and identifies a set of 128 questions that Zero-Shot-CoT answers incorrectly, with an error rate of 21.3% over the full test set. Experiments reveal that when test questions fall into frequent-error clusters in the embedding space, Retrieval-Q-CoT tends to pick multiple wrong demonstrations that share similar mistakes. This leads to high "unresolving" rates, where even Manual-CoT cannot correct the propagated error. The analysis shows that Retrieval-Q-CoT produces a notably higher unresolving rate than Random-Q-CoT, demonstrating that naive similarity-based retrieval can amplify systematic LLM failures rather than mitigate them.

The notion of frequent-error clusters is particularly informative. By clustering questions with Sentence-BERT embeddings and examining error rates per cluster, the paper shows that some clusters exhibit substantially higher error rates than others. This structure reveals that LLM mistakes are not independent across questions but instead tend to concentrate within certain semantic patterns. The paper leverages this observation to argue for diversity-based rather than similarity-based demonstration sampling, which is a clear conceptual contribution.

1.3 Modular Auto-CoT algorithm with principled design choices

The proposed Auto-CoT method is structured as a two-stage pipeline that is simple, modular, and relatively model-agnostic.

Question clustering. In the first stage, all questions in a dataset are embedded using a Sentence-BERT encoder. The embeddings are then partitioned into a predefined number of clusters using k-means. The number of clusters equals the number of desired demonstrations, so exactly one representative question is drawn from each cluster. This design enforces diversity: each cluster roughly corresponds to a distinct semantic pattern in the dataset, and cluster centers serve as prototypes for these patterns. Visualization with PCA indicates that, across ten datasets, clusters align with interpretable patterns in question text.

Demonstration sampling. In the second stage, Auto-CoT samples a representative question from each cluster and generates its reasoning chain using Zero-Shot-CoT. The paper then applies simple but effective heuristics to reduce the risk of low-quality demonstrations. The heuristics encourage shorter questions and shorter rationales by restricting selected demonstrations to questions with at most 60 tokens and rationales with at most five reasoning steps (approximated by newline counts). For arithmetic tasks except AQuA, the answer is required to be non-empty and to appear in the rationale, which reduces answer-rationale mismatches. These design choices explicitly target the error modes identified in Appendix A.1, where rationales with mismatched answers or unnecessarily long reasoning are shown to harm performance.

The impact of these heuristics is quantified in Table 9, which reports the average number of wrong rationales among constructed demonstrations over three runs. For MultiArith, the number of wrong rationales drops from 1.3 to 0.3 out of 8 demonstrations when heuristics are applied. For AddSub, the number reduces from 5.0 to 1.7, and for GSM8K from 3.0 to 1.7. Figure 10 further shows that with heuristics, the error rate among demonstrations can be kept below 20% in seven out of ten tasks. This empirical evidence illustrates that simple, generic rules can substantially improve demonstration quality without manual hand-crafting.

1.4 Systematic ablation and robustness studies

Beyond the main comparison, the paper conducts multiple ablation and analysis studies that clarify why Auto-CoT works.

First, Section 5.5 examines the effect of wrong demonstrations. The study interpolates between In-Cluster Sampling, which intentionally selects multiple demonstrations from frequent-error clusters, and Auto-CoT, which samples from diverse clusters. Figure 6 plots accuracy as a function of the percentage of wrong demonstrations in the demonstration set. For Auto-CoT, accuracy remains relatively stable when one or two demonstrations out of eight are wrong. In contrast, In-Cluster Sampling exhibits a sharp drop in accuracy as the fraction of wrong demonstrations increases. This experiment provides quantitative support for the claim that diversity in demonstrations mitigates the impact of residual errors in Zero-Shot-CoT-generated rationales.

Second, the paper evaluates a more realistic streaming setting in Section 5.6. In this setting, questions arrive in batches of size 30 and Auto-CoT cannot access the entire dataset in advance. The proposed Auto-CoT* procedure bootstraps from an initially empty memory by using Zero-Shot-CoT for the first batch and then applying Auto-CoT on accumulated question-chain pairs for subsequent batches. Figure 7 reports accuracy on MultiArith per batch. As expected, Auto-CoT* and Zero-Shot-CoT coincide on the first batch. From the second batch onward, Auto-CoT* performs comparably to Manual-CoT across batches, indicating that automatically constructed demonstrations remain effective under online constraints.

Third, Section 5.4 demonstrates that Auto-CoT maintains competitive performance when the underlying LLM is switched from GPT-3 to Codex. On MultiArith, Auto-CoT with Codex achieves 93.2%, compared to 96.8% for Manual-CoT and 64.8% for Zero-Shot-CoT. On GSM8K, Auto-CoT reaches 62.8%, outperforming Zero-Shot-CoT at 31.8% and slightly surpassing Manual-CoT at 59.4%. These results show that Auto-CoT is not tied to a single language model and can leverage stronger LLMs when they are available.

2 Key Limitations

2.1 Task and data coverage

Although the paper covers ten benchmarks, the tasks are primarily short-form textual reasoning problems focused on arithmetic, symbolic operations, and multiple-choice commonsense questions. Many of these datasets, such as MultiArith and SingleEq, consist of relatively small-scale, template-like math word problems. There is limited evaluation on long-context reasoning, multi-hop question answering, open-domain knowledge-intensive tasks, or multi-modal settings. As a result, the demonstrated effectiveness of Auto-CoT is strongest for structured, well-defined problems and less clearly established for more complex, noisy, or real-world scenarios.

In addition, the experiments focus on English-language datasets. The study does not explore multilingual or cross-lingual settings, where both question embedding quality and Zero-Shot-CoT behavior may differ significantly. The generalizability of Auto-CoT to other languages therefore remains untested.

2.2 Model and setting limitations

The core experiments rely on a single family of large proprietary models. GPT-3 (text-davinci-002) is used for most evaluations, and Codex is used in a limited set of additional experiments. There is no systematic investigation of smaller LLMs, open-source models, or models with different architectures. The extent to which Auto-CoT requires very large models with strong Zero-Shot-CoT capabilities is therefore not fully quantified.

Furthermore, Auto-CoT assumes access to the full set of evaluation questions when constructing demonstrations. The method clusters all available questions and samples one representative per cluster before evaluation. This setting is closer to a transductive scenario, in which test inputs are known at construction time, rather than a strictly inductive zero-shot regime. Although the streaming extension Auto-CoT* relaxes this assumption, the main reported results in Table 3 still benefit from full-dataset access. The implications of this design choice for fairness in comparison to methods that do not see test questions in advance are not explicitly discussed.

2.3 Evaluation metrics and error analysis

The paper uses exact answer accuracy as the sole evaluation metric. While accuracy is appropriate for many reasoning benchmarks, it does not capture several important aspects of CoT prompting, such as the faithfulness, conciseness, or readability of rationales. The study does not include human evaluation of explanation quality or any automatic measure of rationale consistency. This omission is noteworthy because Auto-CoT explicitly constructs reasoning chains, and some Zero-Shot-CoT-generated rationales, even when leading to correct answers, may contain spurious or logically incorrect steps.

The error analysis, although insightful in identifying frequent-error clusters, remains relatively shallow from a linguistic and cognitive perspective. The paper shows that certain clusters have high error rates and that similarity-based retrieval tends to pick multiple mistakes from these clusters. However, it does not categorize these clusters in terms of underlying reasoning skills, linguistic patterns, or semantic types. As a result, the analysis is more descriptive than diagnostic and offers limited guidance on how to systematically remove or repair these failure modes beyond encouraging diversity.

2.4 Algorithmic and practical limitations

Auto-CoT depends critically on the quality of Sentence-BERT embeddings and k-means clustering. The study visualizes clusters and reports that they exhibit interpretable patterns, but it does not compare alternative clustering algorithms or embedding models in detail. The choice of the number of clusters is treated mainly as a hyperparameter tied to the desired number of demonstrations. There is no exploration of adaptive or data-driven strategies for choosing this number, nor of hierarchical clustering or other structures that might better capture task heterogeneity.

The heuristics used in demonstration sampling, though effective, are relatively ad hoc. Thresholds such as 60 tokens for question length and five steps for rationale length are inspired by manual CoT design guidelines but are not systematically tuned or justified. These thresholds might not extrapolate to tasks with longer descriptions, multi-turn dialogs, or more complex reasoning chains. The rule that the answer appears in the rationale is tightly coupled to arithmetic tasks where the answer is a number at the end of the rationale and may not generalize well to tasks with non-numeric or span-based outputs.

Finally, the paper does not provide computational cost or latency analysis. Auto-CoT requires invoking Zero-Shot-CoT to generate rationales for multiple candidate questions in each cluster, and running clustering over potentially large question sets. For datasets with thousands of questions, the number of LLM calls required to construct demonstrations can be substantial. Without reporting the number of API calls or wall-clock time, it is difficult to assess whether Auto-CoT is practical in cost-sensitive or real-time environments.

3 Technical Bottlenecks

3.1 Quality and reliability of Zero-Shot-CoT rationales

Auto-CoT relies on Zero-Shot-CoT to generate rationales and answers for candidate demonstration questions. However, the paper itself emphasizes that large language models are "still not perfect zero-shot reasoners" and that Zero-Shot-CoT exhibits non-trivial error rates. In MultiArith, the overall error rate is 21.3%, and in some frequent-error clusters the error rate is even higher. These errors directly affect the quality of constructed demonstrations.

The paper partially mitigates this bottleneck through diversity-based sampling and simple heuristics, yet it does not fundamentally address the core technical limitation that LLM-generated rationales can be systematically wrong. In tasks where Zero-Shot-CoT is less reliable than in the studied benchmarks, Auto-CoT may inherit a higher fraction of incorrect demonstrations, which could severely degrade in-context learning.

3.2 Structured failure modes and frequent-error clusters

The discovery of frequent-error clusters indicates that LLM errors exhibit structure in the embedding space: questions in certain regions tend to induce similar mistakes. Retrieval-Q-CoT, which retrieves nearest neighbors by question similarity, is particularly vulnerable to this structure because it tends to pick multiple wrong demonstrations with highly correlated errors. This leads to a high unresolving rate where even strong manual CoT prompts cannot recover correct answers for these test questions.

While Auto-CoT alleviates this issue through cross-cluster sampling, the underlying technical bottleneck remains: the representation space and LLM behavior jointly create regions of concentrated failure. The paper does not propose mechanisms to detect or repair frequent-error clusters directly. As a result, Auto-CoT treats these clusters as sources of potential noise rather than opportunities for targeted model improvement.

3.3 Trade-off between diversity and relevance

Auto-CoT operationalizes diversity by sampling one demonstration from each cluster, regardless of the similarity between the cluster center and a particular test question. This design is motivated by the negative results of similarity-based Retrieval-Q-CoT. However, it introduces a new trade-off between diversity and instance-level relevance.

In principle, demonstrations that are semantically similar to a test question can provide highly informative patterns, provided that their rationales are correct. Auto-CoT prefers global coverage of question patterns over local similarity. Figure 6 shows that accuracy is robust when only a small fraction of demonstrations are wrong, but the paper does not deeply analyze how this robustness changes when similarity is reintroduced under stronger quality control. The design space of combining diversity and similarity in a principled way therefore remains largely unexplored.

3.4 Representation and clustering limitations

The method uses Sentence-BERT for question embeddings and k-means for clustering. These choices are reasonable but introduce technical bottlenecks:

- Sentence-BERT is trained for general-purpose sentence similarity and may not capture all task-specific reasoning features, such as fine-grained arithmetic structure or symbolic patterns.
- K-means assumes roughly spherical clusters in the embedding space, which may not hold for heterogeneous reasoning datasets that contain multiple intertwined templates and skills.

Although Figure 5 suggests that clusters correspond to meaningful patterns, there is no quantitative evaluation of clustering quality tied to downstream CoT performance. The dependence of Auto-CoT on the embedding and clustering choices therefore introduces a potential performance ceiling that is not fully characterized.

3.5 Bootstrapping and error accumulation in streaming settings

The Auto-CoT* extension for streaming settings bootstraps from an initially empty memory by adding question-chain pairs batch by batch. Figure 7 shows that, on MultiArith, Auto-CoT* quickly reaches performance comparable to Manual-CoT after the first batch. However, this behavior heavily depends on the accuracy of Zero-Shot-CoT in early batches and on the robustness of Auto-CoT to noisy demonstrations.

If early batches contain many difficult questions that Zero-Shot-CoT fails on, the resulting memory set may accumulate a non-negligible number of incorrect rationales. The current analysis does not fully explore such adversarial or worst-case scenarios. There is no formal guarantee on the stability of the bootstrapping process when initial demonstrations are substantially noisy.

4 Research Implications

4.1 Reframing CoT prompting as automatic demonstration construction

The primary implication of the paper is that CoT prompting can be reframed from manual prompt engineering to automatic demonstration construction. Manual-CoT requires hand-designed demonstrations with human-written rationales, which is time-consuming and task-specific. Auto-CoT shows that demonstrations can be constructed automatically from unlabeled questions and Zero-Shot-CoT rationales while matching or exceeding Manual-CoT performance on a suite of benchmarks.

This result suggests that the bottleneck in applying CoT prompting to new tasks may shift from human effort to computational resources for automatic demonstration generation. In practice, this can substantially lower the barrier to deploying CoT prompting in new domains, provided that the cost of LLM calls is acceptable.

4.2 Insights into in-context learning dynamics

The comparison between Retrieval-Q-CoT, Random-Q-CoT, and Auto-CoT offers non-trivial insights into how in-context learning operates in large language models. The failure of similarity-based Retrieval-Q-CoT in the presence of frequent-error clusters suggests that in-context learning is sensitive not only to the surface similarity between demonstrations and test questions but also to the correctness and variability of demonstration rationales.

The success of diversity-based sampling indicates that covering a broad range of question types and reasoning skills can be more beneficial than focusing narrowly on nearest neighbors in embedding space. This observation connects CoT prompting to classical ideas in machine learning, such as coverage, skill composition, and robustness to label noise. It also challenges simplistic interpretations of in-context learning as nearest-neighbor retrieval, indicating that more nuanced mechanisms may be at play.

4.3 Structured error patterns as a tool for model diagnosis

The frequent-error cluster phenomenon observed in Section 3.2 has broader implications for model diagnosis and debugging. If errors tend to cluster in specific regions of semantic space, then identifying and characterizing these clusters can provide a structured way to understand model weaknesses. Rather than treating failures as isolated instances, one can view them as manifestations of missing or mislearned skills.

In this context, Auto-CoT can be interpreted as a first step toward skill-aware prompting, where demonstrations are selected to cover diverse skills while avoiding concentrated failure regions. Future diagnostic tools might build on similar clustering analyses to locate and address systematic reasoning gaps.

4.4 Data-free or weakly supervised adaptation

Auto-CoT constructs demonstrations without access to ground-truth labels or human-written rationales. It uses Zero-Shot-CoT outputs as pseudo-labels and pseudo-rationales. This positions Auto-CoT within the broader landscape of data-free or weakly supervised adaptation methods, in which models adapt to new tasks by leveraging their own predictions or self-generated data.

The empirical success of Auto-CoT implies that, for certain reasoning tasks, pseudo-rationales generated by a strong LLM are accurate enough to serve as effective teaching signals for in-context learning. This has implications for scenarios where labeled data are scarce or expensive but powerful LLMs are available.

4.5 Reliability and safety considerations

The finding that Auto-CoT remains effective even when a small fraction of demonstrations are wrong has two-sided implications. On the positive side, it indicates that in-context learning with CoT prompting is robust to a moderate amount of label noise in demonstrations. On the negative side, it suggests that LLMs can still perform well even when exposed to rationales that contain systematic errors, which may be problematic in safety-critical applications where incorrect but plausible reasoning chains could mislead users.

The paper does not explicitly address these safety aspects, but its results highlight the need for evaluation metrics and methods that assess not only final accuracy but also the reliability and truthfulness of intermediate reasoning steps.

5 Potential Research Directions

5.1 Quality-controlled demonstration generation

A natural extension of Auto-CoT is to introduce explicit quality control mechanisms for generated rationales and answers. Possible directions include:

- Integrating external verifiers, such as symbolic math solvers for arithmetic tasks or program executors for code-related tasks, to filter out demonstrations with incorrect final answers.
- Using self-consistency, in which multiple Zero-Shot-CoT samples are generated for each candidate question and only consistent rationales are retained as demonstrations.
- Estimating confidence scores for rationales and discarding demonstrations whose rationales appear uncertain or internally inconsistent.

Such methods could reduce the residual error rate among demonstrations beyond what simple heuristics achieve.

5.2 Learning demonstration selection policies

Instead of relying on fixed clustering and heuristic sampling, future work can frame demonstration selection as a learning problem. A selection policy can be trained to maximize downstream accuracy given a limited demonstration budget. Approaches include reinforcement learning over demonstration subsets, gradient-based meta-learning that optimizes demonstration sets on validation tasks, or differentiable surrogate objectives that correlate with in-context performance.

This line of research could systematically explore the trade-off between diversity and similarity, learning to choose demonstrations that are both varied and relevant to incoming test questions.

5.3 Task-specific embedding and clustering

The current method uses off-the-shelf Sentence-BERT embeddings and standard k-means clustering. Task-specific representation learning could improve cluster quality and, in turn, demonstration coverage. Possible strategies include:

- Training or fine-tuning sentence encoders on the specific reasoning tasks, optimizing for separation between different reasoning patterns rather than general sentence similarity.
- Exploring alternative clustering algorithms, such as spectral clustering, density-based clustering, or hierarchical clustering, which may better capture complex structure in question space.
- Developing criteria for automatically selecting the number of clusters based on data characteristics or validation performance.

Such improvements may increase the effectiveness and robustness of Auto-CoT across diverse domains.

5.4 Extending to complex and multi-modal tasks

Auto-CoT is evaluated on single-turn textual reasoning tasks. Extending the framework to more complex scenarios is an important direction. Examples include:

- Multi-hop question answering where the model must combine information from multiple sentences or documents.
- Dialog-based reasoning where the context includes conversational history and user-specific information.
- Vision-language tasks such as visual question answering, where questions are paired with images and reasoning chains may need to reference visual evidence.

These settings would require richer question representations, modified clustering strategies, and potentially multimodal CoT demonstrations.

5.5 Richer evaluation metrics and analysis tools

The focus on accuracy leaves several aspects of CoT behavior unexplored. Future work can develop and adopt metrics that capture:

- The logical coherence and faithfulness of rationales.
- The calibration of model predictions, particularly when rationales are incorrect.
- The diversity and coverage of reasoning skills represented in demonstration sets.

Combining these metrics with clustering-based analyses of frequent-error regions may lead to more informative evaluations and clearer guidance for model improvement.

5.6 Theoretical analysis of noise tolerance in in-context learning

The empirical finding that Auto-CoT remains robust to a small fraction of wrong demonstrations invites theoretical investigation. Questions include:

- Under what conditions on the distribution of demonstrations and errors does in-context learning remain stable?
- How does the fraction and structure of label noise in demonstrations affect the expected performance of LLMs under CoT prompting?

- Can formal connections be drawn between CoT in-context learning and classical results on learning with noisy labels?

Answering these questions would deepen understanding of why Auto-CoT works and provide principled guidelines for acceptable levels of demonstration noise.

6 Conclusion

The paper on Automatic Chain-of-Thought Prompting in Large Language Models presents a clear and well substantiated contribution: automatic construction of CoT demonstrations can match or surpass manually designed prompts on a broad suite of reasoning benchmarks. The methodological strengths include a comprehensive experimental design, thorough analysis of failure modes of naive retrieval-based prompting, a modular and intuitive Auto-CoT algorithm, and systematic ablation studies that support the central claim that diversity in demonstrations is crucial.

At the same time, the study exhibits several limitations. The evaluation focuses on short-text reasoning tasks in English, relies on large proprietary LLMs, and uses accuracy as the sole metric. The algorithm depends on specific embedding and clustering choices, relies on heuristics for demonstration filtering, and lacks detailed cost analysis. These limitations highlight technical bottlenecks in Zero-Shot-CoT reliability, structured failure regions, and the trade-off between diversity and relevance.

The most promising research directions include quality-controlled demonstration generation, learning-based demonstration selection, task-specific representations and clustering, extensions to more complex and multimodal tasks, richer evaluation metrics, and theoretical analysis of noise tolerance in in-context learning. Together, these avenues build on the insights of Auto-CoT and point toward a more principled understanding and deployment of chain-of-thought prompting in future reasoning systems.